EC2

What's EC2

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud. It is designed to make web-scale cloud computing easier for developers.

Important If you are shutting down instance from OS its not stop costing you . You need to shutdown instance from terminal

EC2 Options

- **On demand:** You pay for computing capacity by per hour or second with no commitment
- **Reserved Instance** (RI): Provide a significant discount (up to 75%) compared to On-Demand pricing and provide a capacity reservation when used in a specific Availability Zone. You have to enter a contract.
    - (Standard, Convertible, or Scheduled)
    - capacity reservation on 1 or 3 year contract terms
- **Spot**: Amazon EC2 Spot instances allow you to request spare Amazon EC2 computing capacity for up to 90% off the On-Demand price.
    - If you terminate an instance, you will pay for the complete hour.
    - If Amazon terminates the instance, you won't pay for the complete hour.
    - The following are the possible reasons that Amazon EC2 might interrupt your Spot Instances:
        - Price – The Spot price is greater than your maximum price.
        - Capacity – If there are not enough unused EC2 instances to meet the demand for Spot Instances, Amazon EC2 interrupts Spot Instances. The order in which the instances are interrupted is determined by Amazon EC2.
        - Constraints – If your request includes a constraint such as a launch group or an Availability Zone group, these Spot Instances are terminated as a group when the constraint can no longer be met.
- **Dedicated Hosts**: Is a physical server with EC2 instance capacity fully dedicated to your use.
    - Dedicated Hosts, physical EC2 server, useful for server-bound software licenses

What's EBS

Provides persistent block storage volumes for use with Amazon EC2 instances in the AWS Cloud. EBS is automatically replicated in a specific AZ

Amazon EBS Volume Types

SSD

SSD are good for random access.

- General Purpose SSD (GP2): General purpose SSD volume that balances price and performance. General purpose SSD. For web applications // most use cases
    - 3 IOPS per GB
    - 1GB to 16 TB

    - 16000 IOPs
- Provisioned IOPS SSD (io1): Designed for IO intensive, For critical high performing databases.
    - 4 to 16 TB
- 64000 IOPs

HDD (Magnetic)

Are great for sequential access (processing log files, bigdata work flows as an example).

- Throughput Optimized HDD (st1): Magnetic disk, this can't be a boot volume (root volume). For Big Data
    - 500 GB to 16 TB
    - 500 IOPs
- Cold HDD (sc1): Lower cost storage, like file servers, can't be a boot volume.infrequently accessed data
    - 500 GB to 16 TB
    - 250 IOPs
- Magnetic (standard): Lowest cost per gigabyte of all EBS. It's bootable and it's from the previous storage generation.
    - 1 GB to 1 TB
    - 40 to 200 IPs

Exam, Maximum throughput EBS 1750 MB and 80,000 IO

EBS Volume and Snapshot Encryption

- DEK , CEK

EBS Optimization, Enhanced Networking,

- EBS Optimization create a separate path for ec2 from networking path
- Enhanced Networking physical network card act like fake multiple network card

RAID Arrays using EBS

As you would do in a bare-metal server, you can also create RAID arrays within AWS EC2 boxes using EBS volumes.

- Remember that the RAID available in AWS is only software
- You can have two types of RAID:
    - RAID 0 – splits ("stripes") data evenly across two or more disks. When I/O performance is more important than fault tolerance; for example, as in a heavily used database (where data replication is already set up separately). You can use RAID 0 configurations in scenarios where you are using heavy databases with perhaps mirroring and replication.
    - RAID 1 – consists of an exact copy (or mirror) of a set of data on two or more disks. When fault tolerance is more important than I/O performance; for example, as in a critical application. With RAID 1 you get more data durability in addition to the replication features of the AWS cloud.
    - RAID 5 and RAID 6 are not recommended for Amazon EBS because the parity write operations of these RAID modes consume some of the IOPS available to your volumes. Depending on the configuration of your RAID array, these RAID modes provide 20-30% fewer usable IOPS than a RAID 0 configuration. Increased cost is a factor with these RAID modes as well; when using

identical volume sizes and speeds, a 2-volume RAID 0 array can outperform a 4-volume RAID 6 array that costs twice as much.

Remember also that you can create snapshots of your RAID arrays

RAID0 vs RAID1 —

- RAID1 is used for mirroring, high-availability and redundancy.
- RAID0 is used for higher performance, it can combine multiple disk drives together.

Launch an EC2 Instance

- Termination protection is turned off by default.
- The EBS root volume by default is deleted at termination.
- Default AMI's (provided by Amazon) cannot be encrypted.
- Additional volumes can be encrypted.

Security groups

What's a security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic.

- Security Group (ENI elastic network interface)
- All Inbound traffic is Blocked by Default.
- All Outbound traffic is Blocked by Default.
- All security groups changes are applied immediately.
- Security groups are stateful. For example, if you allow the request to come in, automatically responses can go out even if you don't have anything on the outbound section of your security group.
- You can specify only allow rules, not deny rules.
- You can have many EC2s within a security group and multiple security groups attached to EC2s
- You cannot block sepcific IP addresses (use network ACLs instead)
- Import Q : 5 Security group per ENI
- Security is stateful

EBS Volumes & Encrypt Root Device Volume

- Instances and Volumes MUST be in the same AZ
- Snapshots exists in S3.
- Snapshots are a point in time copies of Volumes.
- Snapshots are incremental (only differences are saved).
- *You can't delete a snapshot of an EBS volume that is used as the root device of a registered AMI.*
- You can change EBS volumes size and type on the fly.
- To move an EC2 volume from one AZ/Region to another, take a snap or an image of it, then you can copy them to the new AZ/Region.
- Snapshots of encrypted volumes are encrypted automatically
- Volumes restored from an encrypted snapshot will be encrypted as well.
- You can share snapshots only if they are not encrypted, these snapshots can be made public.

AMI Types

- EBS: Amazon EBS provides durable, block-level storage volumes that you can attach to a running instance
    - EBS takes less time to provision.
    - EBS volumes can be kept once the instance is terminated.
- Instance Store / Ephemeral storage: This storage is located on disks that are physically attached to the host computer
    - Instance Store Volumes can't be stopped, if the host fails, you lose data.
    - You can reboot the instance without losing data.
    - You can not detach Instance Store Volumes.
    - Instance store volumes cannot be kept once the instance is terminated.

Elastic Load Balancers

What's an Elastic Load Balancer

Elastic Load Balancing automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances, and others.

Types of loadbalancers

- Application Load balancers: Best for load balancing HTTP & HTTPS traffic. They operate at layer 7.
- Network Load Balancer: Loadbalancing TCP traffic where extreme performance is needed. They operate at layer 4.
- Classic Load Balancer: Legacy ELB, mostly operate at layer 4, but can go up to 7.

X-Forwarded-For

(XFF) The HTTP header field is a common method for identifying the originating IP address of a client connecting to a web server through an HTTP proxy or load balancer.

Load Balancers

- You must set up health checks in order to let the load balancer to understand if the instances are up and running
- Load balancers won't expose any IP, instead they expose DNS names.

Patching an AMI for an auto scaling group, the procedure is —

1. Create an image out of the main patched EC2 instance
2. Create a new launch configuration with new AMI ID
3. Update auto scaling group with new launch configuration ID.

Note that AMI ID is set during creation of launch configuration and cannot be modified, so we have to create a new launch configuration.

Default metric types for a load balancer —

1. Request count per target.

2. Average CPU utlization.
3. Network in.
4. Network out.

Monitoring Application Load Balancers —

1. Cloudwatch metrics
2. Access logs
3. Request tracing
4. Cloudtrail logs.

Adding lifecycle hooks to ASGs put instances in wait state before termination. During this wait state, we can perform custom activities. Default wait period is 1 hour.

ASG Dynamic Scaling Policies —

- Target tracking scaling. The preferred one to use, this should be the first one we should consider.
- Step scaling
- Simple scaling

If you are scaling based on a utilization metric that increases or decreases proportionally to the number of instances in an Auto Scaling group, we recommend that you use target tracking scaling policies. Otherwise, we recommend that you use step scaling policies.

The ELB service does not consume an IP address, it's the nodes that cosume one IP address each.

Auto-scaling ensures —

- Fault tolerance
- Availability

ELBs can manage traffic within a region and not between regions.

For unstable scaling behavior, that is scaling multiple times frequently, the following things can be done —

- Increasing auto-scaling cooldown timer value would give scaling activity sufficient time to stabilize.
- Modify the cloudwatch alarm period that triggers scaling activity.

Default cooldown period is 300 seconds.

Port based routing is supported by Application Load Balancer.

Network Load Balancer can be used to terminate TLS connections. For this, NLB uses a security policy which consists of protocols and ciphers. The certificate used can be provided by AWS Certificate Manager.

Connection draining enables the load balancer to complete in-flight requests made to instances that are de-registering or unhealthy.

ASG termination policy —

1. Oldest launch configuration.

2. Closest to next billing hour.
3. Random.

Load balancer does not create or terminate instances, that's done by auto scaling group.

CloudWatch

- Enchance Monitoring go to 1 min default is 5 min
- Standard monitoring is 5 min and * Detailed monitoring is 1 min (you will be charged for it)
- Dashboard used to visualize what's happening with your AWS environment.
- Alarms can be set to notify when a specific threshold is hit
- Events can be used to perform actions when state changes happen in your AWS resources.
- Logs can be aggregated in a single place to better troubleshoot. Remember that you need to install an agent on the EC2 instance.
- By default, Matrics on EC2 instances are: CPU related, Disk related, Network related and Status check related.

Remember that:

- CloudWatch: is for monitoring resources
- CloudTrail: is for auditing

IAM Roles with EC2 - Lab

- Roles avoid you to store credentials inside EC2 instances in order to communicate with other AWS services
- Roles can be assigned after an EC2 instance it has been provisioned
- Roles are universal, you can use them in any region.

EC2 Instance Metadata

Within your local instance run:

```
http://169.254.169.254/latest/meta-data
```

This will give you the top level metadata items, something like this.

```
[ec2-user ~]$ curl http://169.254.169.254/latest/meta-data/
ami-id
ami-launch-index
ami-manifest-path
block-device-mapping/
events/
hostname
iam/
instance-action
instance-id
instance-type
local-hostname
```

```
local-ipv4
mac
metrics/
network/
placement/
profile
public-hostname
public-ipv4
public-keys/
reservation-id
security-groups
services/
```

If you want to access a specific item, simply add it at the end of your request:

```
[ec2-user ~]$ curl http://169.254.169.254/latest/meta-data/ami-id
ami-12345678
```

There is also one URL for user data

```
[ec2-user ~]$ curl http://169.254.169.254/latest/user-data/
```

Launch configurations and Autoscaling Groups

- A launch configuration is an instance configuration template that an Auto Scaling group uses to launch EC2 instances
- An Auto Scaling group contains a collection of Amazon EC2 instances that are treated as a logical grouping for the purposes of automatic scaling and management.

Autoscaling group will automatically spread evenly on the number of instances across the AZ you selected once you configured it. So 3 AZ with 3 as group size, means 1 box in each AZ

Placement Groups

You can launch or start instances in a placement group, which determines how instances are placed on the underlying hardware. When you create a placement group, you specify one of the following strategies for the group:

- **Cluster** – clusters instances into a low-latency group in a single Availability Zone. physically close best performance
- **Spread** – spreads instances across underlying hardware and can spread in multiple Availability Zones.Spread. Maximum number of instances in an AZ is 7
    - Individual Instances in each rack
    - large environment
- **Partition** – spreads instances across logical partitions, ensuring that instances in one partition do not share underlying hardware with instances in other partitions.
    - small application , email servers

Some notes about placement groups:

- The name you specify for a placement group must be unique within your AWS account.
- Only specific types of instances can be launched in a placement group.
- You can't merge placement groups.
- You can't move an existing instance into a placement group.
- If the exam refers to placement groups, without mentioning which type, it's most probably talking about the Cluster ones since those are the old ones.
- Placement group names must be unique within your AWS account
- Placement groups cannot be merged
- Existing instances cannot be moved into a placement group (create an AMI instead)
- Cluster Placement Group
    - Grouping of certain instances within a single AZ (cannot span multiple AZs)
    - Recommended for low network latency and/or high network throughput
- Spread Placement Group
    - Group of instances that are each placed on distinct underlying hardware
    - Recommended for a small number of critical instances that should be kept separate
    - Can span multiple AZs

EFS

Amazon EFS provides scalable file storage for use with Amazon EC2. You can create an EFS file system and configure your instances to mount the file system

- Supports NFSv4.
- You only pay for the storage you use.
- Scale up to petabytes.
- Can support thousands of concurrent connections.
- Data is stored across multiple AZ.
- Read after write consistency.
- You need to make sure that the EC2 instance that needs to connect with the EFS volume, is associated with the same security group you have on the EFS volume.
- You can assign permissions at the file level and at the folder level.
- 2+ instances can share an EFS volume
- Grows and shrinks automatically as files are added or removed
- Petabyte-scale storage that supports thousands of NFSv4 connections
- Pay for the storage you use (no pre-provisioning required)
- Data is stored across multiple AZs within a region
- Read after Write consistency
- NFS port 2049 for EFS.
- Throughput mood : Bursting and provisioning
- Storage class: Standard and infrequently

Lambda

AWS Lambda lets you run code without provisioning or managing servers. You pay only for the compute time you consume - there is no charge when your code is not running. Lambda basically is based on triggers.

You can use lambda in two ways:

- Event-driven based: Lambda runs your code based on events, such, new file on S3 or a new alarm on cloudwatch
- Compute-service based: Lambda runs your code based on HTTP requests using an API Gateway or API calls made using AWS SDKs.
- Lambda is FaaS
- Statless

AWS lambda, supports: C#, Node.js, Python, and Java

Lambda is charged as follows:

- First 1M requests per month are free. $0.20 PER 1M requests thereafter
- Duration: You are charged for the amount of memory you allocate on your functions. First 400,000 GB-seconds per month, up to 3.2M seconds of computing time, are free.
- Your functions can't go over 15 minutes in run-time.

Lambda functions can be run within a private VPC.

Lambda can read events from —

- Amazon Kinesis
- Amazon DynamoDB
- Amazon Simple Queue Service

Services that can invoke Lambda functions —

- Elastic Load Balancing (Application Load Balancer)
- Amazon Cognito
- Amazon Lex
- Amazon Alexa
- Amazon API Gateway
- Amazon CloudFront (Lambda@Edge)
- Amazon Kinesis Data Firehose
- Amazon Simple Storage Service
- Amazon Simple Notification Service
- Amazon Simple Email Service
- AWS CloudFormation
- Amazon CloudWatch Logs
- Amazon CloudWatch Events
- AWS CodeCommit
- AWS Config

AWS CodePipeline and AWS OpsWorks can't invoke lambda functions.

For failures we can configure lambda to send non-processed payloads to SQS Dead letter queue. Then we can configure SNS to send a notification if we want. Lambda does not have an in-built mechanism for notification upon failure.

A policy on a role defines which API actions can be made on the target, it does not define whether the source can access the target or not.

Each lambda function has an ephemeral storage of 512 MB in the `tmp` directory.

AWS CloudWatch rule can be configured to trigger a lambda function. While configuration, the following can be used as input to the target lambda function —

- Matched event
- Part of the matched event
- Constant (JSON text)

The following CloudFront events can trigger lambda function —

- Viewer request
- Viewer response
- Origin request
- Origin response

Lambda function update has eventual consistency. Which means, for a brief window of less than a minute, it may execute either the old version or the new version.

We can use alias versions to point to another version. This can enable easier upgradation from the viewpoint of a consumer.

Limits —

- Function memory allocation: 128 MB to 3008 MB, in 64 MB increments.
- Function timeout: 900 seconds.
- Deployment package: 50 MB * 5 layers.
- `tmp` directory storage: 512 MB.

To grant cross-account permission to a function, we have to modify the function policy, not the execution role policy.

The console doesn't support directly modifying permissions in a function policy. You have to do it from the CLI or SDK.

If we run lambda functions inside a VPN, they use subnet IPs or ENIs. There should be sufficient ones otherwise it will get throttled.

ENI capacity = Projected peak concurrent executions * (Memory in GB / 3 GB).

The lambda console provides encryption and decryption helpers for encryption of environment variables.

By default, the a KMS default service key is used for encryption, which makes the information visible to anyone who has access to the lambda console. For further restriction, create a custom KMS key and use that to encrypt.

CloudWatch metrics for Lambda —

- Invocations
- Errors
- Dead Letter Error
- Duration
- Throttles

- IteratorAge
- ConcurrentExecutions
- UnreservedConcurrentExecutions

We can get the function version within the function using —

- `getFunctionVersion` from the Context object.
- `AWS_LAMBDA_FUNCTION_VERSION` environment variable.

Lambda Retry upon Failure Behavior —

- Event sources that aren't stream-based
    - Synchronous invocation — Returns error with status code 200. Includes FunctionError field and X-Amz-Function-Error header.
    - Asynchronous invocation — Retry twice, then sent to Dead Letter Queue.
- Poll-based and stream-based event source (Kinesis or DynamoDB) — Lambda keeps retrying until the data expires. The exception is blocking, this ensures the data are processed in order.
- Poll-based but not stream-based event source (SQS) — On unsuccessful processing or if the function times out of the message, it is returned to the queue, and ready for further reprocessing after the visibility timeout period. If the function errors out, it is sent to Dead Letter Queue.

Lambda traffic shifting —

- Canary
- Linear
- All at once

Make sure you have a look at what CORS is.

ECS

Launch types —

- Fargate
- EC2

All types of instances, i.e. on-demand, spot and reserved can be used with ECS.

Docker containers and ECS are particularly suited for batch job workloads as they can get embarassingly parallel.

Amazon ECS enables you to inject sensitive data into your containers by storing your sensitive data in either —

- AWS Secrets Manager secrets
- AWS Systems Manager Parameter Store parameters

S3

What's S3

- Object-based storage: you can save only object
- you can't install an OS (In this case you need block-based storage).
- No storage Limits.unlimited storage
- Files are stored in Buckets (a folder in a cloud).
- S3 is a universal namespace, the name must be unique globally. So you cannot have the same name as someone else.
- When you upload an object in S3 you get an HTTP 200 OK code back
- newly created S3 bucket, everything // every additional option is turned off by default. Also, no bucket policy exists
- Transfer Acceleration uses edge locations and Amazon's backbone network (CloudFront)
- Default maximum number of buckets per region: 100
- You can write objects directly to an edge location (for instance, this is useful if you are letting users of an app upload files directly to S3 and you want to reduce latency world wide).
- Presigned URLs allow access to objects on a temporary basis. They are created, and the bearer of the URL has the same level of authorization as the creator

S3 bucket endpoints formats —

1. http://bucket.s3.amazonaws.com
2. http://bucket.s3.aws-region.amazonaws.com
3. http://s3.amazonaws.com/bucket
4. http://s3.aws-region.amazonaws.com/bucket

Object sizes — S3 can store objects of size 0 bytes to 5 TB. A single PUT can transfer 5 GB max. For files larger than 100MB, multipart upload is recommended.

Increasing performance in S3 —

- If workload is mainly GET requests, integrate Cloudfront with S3.
- If workload consists of PUT requests, use S3 transfer acceleration.

AWS S3 performance —

- 3,500 requests per second to add data
- 5,500 requests per second to retrieve data

S3 does not support `OPTIONS`, `CONNECT` and `TRACE` methods.

S3 event notification targets —

- SQS
- SNS
- Lambda

Data Consistency for S3

- **Read after write consistency:**
  - `PUT` (New files)
- **Eventual consistency:**
  - `PUT` (updates)
  - `DELETE`
  - overwriting and deletes (this means it can take some time to propagate)
- S3 is spread across multiple AZ's

Components

- S3 is an object. Objects consist of:
  - Key (name of the object)
  - Value (data)
  - Version ID (Used on versioning)
  - Metadata (a set of data that describes and gives information about the object data.)
  - Subresources:
    - Access Control List (Decide who can access files)
    - Torrent (Not an exam topic)

Basics

- S3 SLA: 99.9% availability
- S3 is built for 99.99%
- S3 guarantees 11x9s (99.999999999) durability for S3 information.
- Tiered Storage (classes) available
- You can have lifecycle management
- Versioning
- Supports multi-part upload
- Encryption
- Access control (permissions on single files) and bucket policies (permissions on buckets)

S3 Storage Tiers

- **S3 standard**: 99.9% availability and 99.999999999% (11 x 9s) durability (it sustains the loss of 2 facilities concurrently)
- **S3 IA**: (Infrequently Accessed): For data that is accessed less frequently, but needs rapid access. You are charged a retrieval fee per GB retrieved.
  - Min object size: 0bytes, except for S3 IA that has a min size of 128kb.
- **S3 One Zone IA**: Like S3 IA but data is stored only in one AZ
- **S3 Intelligent Tiering**: Tiering uses ML to move objects around storage classes, Designed to optimize cost
- **Glacier**: Most cheap, used for archival only.
  - Expedited: few minutes for retrieval
  - Standard: 3-5 hours for retrieval
  - Bulk: 5-12 hours for retrieval
  - It encrypts data by default
  - Regionally availability
  - Designed with 11x9s durability, like S3

- **S3 Glacier Deep** Archive for cheapest storage with a 12 hour retrieval time

Class	Durability	Availability
STANDARD	99,999999999%	99,99%
STANDARD_IA	99,999999999%	**99,9%**
ONEZONE_IA	99,999999999%	**99,5%**
GLACIER	99,999999999%	99,99%*
REDUCED_REDUNDANCY	**99,99%**	99,99%

Charges

S3 is charged for:

- Storage
- Requests
- Storage management pricing
- Data Transfer Pricing
- Transfer acceleration (it's using CloudFront the AWS CDN) using edge locations

S3 Encryption

- Encryption in transit using SSL/TLS
- Encryption at rest, server side
    - S3 Managed Keys (SSE-S3), managed by S3
    - AWS Key Management Service, Managed Keys (SSE-KMS), managed by KMS and the user
    - Server Side Encryption with Customer Provided Keys (SSE-C)
- Encryption at rest, client side

S3 Version Control

- Once you enable versioning, you can't disable it, you can only suspend it. A way of disabling it is to delete the bucket and re-create it
- Every time you update an object, it will become private by default.
- It integrates with Lifecycle rules.
- You pay for each version you have

- Delete an object:
  Once you delete a file inside a versioned bucket, you don't delete the file, you simply add a Delete Marker (this basically creates a new version of the object) If you delete the version with the Detele Marker you will basically restore the object.
  If you want to permanently delete the object, you have to delete all the Versions of the object.
  You can optionally add another layer of security by configuring a bucket to enable MFA Delete
  More info about versioning: ObjectVersioning and Versioning

S3 Cross region replication

- Regions must be unique
- Files in an existing bucket are not replicated automatically
- In order to replicate the existing objects, you need to do a `cp` using the aws cli:
  `aws s3 cp --recursive s3://alessio-casco-versioning s3://alessio-casco-versioning-replica-sydney`
- If you delete an object in the primary bucket, the delete action and markers won't be done or replicated in your remote bucket, this is a security function. Only creations and modifications are replicated to the bucket in the other regions NOT the delete
- You can't replicate over multiple buckets, the maps are always 1-to-1
- Replication delete is not possible delete marker is not going to replicate

S3 Security & Encryption

- You can configure S3 to create access logs for requests made to the S3 bucket
- Access control for buckets:
  - Bucket policies: Permission bucket wide
  - Access control list: Permissions that can be applied to the single object
- Encryption:
  - In transit: from to your bucket, HTTPS for example
  - At rest:
    - Server-side encryption:
      - S3 Managed Keys: SSE-S3 (Keys are managed by S3)
      - Key Management Service: SS3-KMS the customer manages the keys
      - Server-side encryption: Here you manage the keys, and Amazon manage the writes
  - Client-side Encryption: You encrypt the data and you upload it encrypted to S3

CloudFront

What's a CDN

- A content delivery network or content distribution network is a geographically distributed network of proxy servers and their data centres
- Key terminology about CloudFront:
  - Edge Location: Is the location where the content is cached (separate from AWS AZ's or regions) Be aware that you can also write on edge locations, is not ready.
  - Invalidating (erasing) the cache costs money.

        - Origin: Is the source of the files the CDN will distribute. An origin can be an EC2 instance, an S3 bucket, an Elastic Load Balancer or Route53, you can also have your own origin, it is not mandatory that is within AWS.
        - Distribution: Is the name AWS calls CDN's.
            - You can Have two types: Web that is for generic web contents and RTMP that is for video streaming
            - TTL: time to live of the cached object.
    - Web distribution typically for websites
    - RTMP (real-time message protocol) distribution used for Adobe Flash media streaming
    - You can clear old cached objects for a fee (invalidating the cache)
    - Restrict access using signed URLs or cookies (e.g. premium Netflix users)
    - Edge location are not just Read only , we can write to them too
    - Objects are cached for TTL
    - You can restrict your content via signed URL
    - Regional cache , its also having another copy other then edge location
    - origin access identities (OAI) allow restriction of an S3 bucket to accept connections only from CloudFront distributions

\

Amazon Storage

Amazon Storage Gateway

What's an Amazon Storage Gateway: AWS Storage Gateway connects an on-premises software appliance with cloud-based storage to provide seamless integration with data security features between your on-premises IT environment and AWS storage infrastructure.

- File Gateway: For flat files, stored directly in S3. You can NFS Mount points
- Volume gateway (iSCSI): Block-based storage
    - Store volume (you keep all your data on prem)
        - Primary data locally and backup to AWS
        - Low latency because your volume is on locally
        - Store to EBS as Snapshot volume
        - Off-site backup
    - Cached Volumes (you keep only the most recent data on prem) Tape Gateway (VTL): Virtual tapes
        - Cached volume only most actively used data
- Connects on-prem software appliance with cloud-based storage
- Available as physical hardware or a VM image for ESXi or Hyper-V
- File Gateway (NFS), flat files are stored as objects in S3 buckets and accessed through a network file system (NFS) mount point
- Volume Gateway (iSCSI), stored or cached disk volumes that can be backed up as point-in-time snapshots in EBS
    - Stored volumes let you store primary data locally which is asynchronously backed up to S3 as EBS snapshots from 1GB - 16TB
    - Cached volumes (1GB - 32TB in size) use S3 for primary data storage but frequently accessed data is retained locally in the storage gateway
- Tape Gateway (VTL), virtual tape library for archiving

offers 2 main volume modes:**Cached volumes**: stores files in the cloud and keeps a local cache to speed up reads

- **Stored volumes**: optimized for low latency, stores files locally and asynchronously sends back up point-in-time snapshots to S3.

Snowball

Import Export is still available and was the first version of snowball, you used to ship your drives to AWS

Snowball is (an appliance) a petabyte-scale data transport solution that uses devices designed to be secure to transfer large amounts of data into and out of the AWS Cloud

- Petabyte-scale (50TB or 80TB) data transport solution to transfer to and from AWS

Snowball edge: is a 100TB data transfer device with onboard storage-computer capabilities. It's like an AWS DC in a box

Snowmobile: AWS Snowmobile is an Exabyte-scale data transfer service used to move extremely large amounts of data to AWS. A truck full of disks basically.

- Snowmobile is exabyte-scale (up to 100PB) for libraries, repositories, and migrations

S3 Transfer Acceleration

Instead of uploading files directly to your S3 bucket, you can use the AWS edge network. Using a specific URL, you upload the file to your local edge and then the file will be uploaded to S3 an example or URL: alessio-casco-accelerate.s3-accelerate.amazonaws.com

Static Website Using S3

- You can use bucket policies to make entire S3 buckets public
- You can use S3 to host only static websites.
- S3 Scales automatically to meet demand

Cors

Cors is security measurement allowing a web application running in one domain to reference in another domain

https://docs.aws.amazon.com/AmazonS3/latest/dev/cors.html

VPC

What's a VPC

Think of a VPC as a virtual data centre in the cloud.

What are the components of Amazon VPC

- A Virtual Private Cloud: A logically isolated virtual network in the AWS cloud. You define a VPC's IP address space from ranges you select.
- Subnet: A segment of a VPC's IP address range where you can place groups of isolated resources.
- Internet Gateway: The Amazon VPC side of a connection to the public Internet.
- NAT Gateway: A highly available, managed Network Address Translation (NAT) service for your resources in a private subnet to access the Internet.
- Virtual private gateway: The Amazon VPC side of a VPN connection.
- Peering Connection: A peering connection enables you to route traffic via private IP addresses between two peered VPCs.
- VPC Endpoints: Enables private connectivity to services hosted in AWS, from within your VPC without using an Internet Gateway, VPN, Network Address Translation (NAT) devices, or firewall proxies.
- Egress-only Internet Gateway: A stateful gateway to provide egress only access for IPv6 traffic from the VPC to the Internet. Only outgoing Ipv6 to internet
- Cant not use IPv6 with Nat gateway
- you can have multiple VPC in a region (default up to 5).
- You can have 1 internet gateway per VPC.
- 1 Subnet = 1 AZ
- Security groups are Stateful, instead, Network ACLs are stateless.

IPv6

- IPv6 not enabled by default
- need to enable it on vpc level
- IPv6 CIDR /56 cant adjust or edit
- IPv6 the OS configure public address via DHCP6
- No private or public concept
- No vpn support

ARP (Address resolution protocol which resolve IP address to mac)

Default VPC

- Amazon provides a default VPC to immediately deploy instances.
- All Subnets in default VPC have a route out to the internet.

VPC Peering

- You can peer one VPC to another VPC using private IP subnets.
- You can peer VPC's with others AWS accounts as well as with other VPC's in the same account.

How to VPC Peering

- Overlapping CIDR Blocks is not supported: You can't connect two VPC's that have the same CIDR.
- Transitive Peering is not supported:
  You have a VPC peering connection between VPC A and VPC B (pcx-aaaabbbb), and between VPC A and VPC C (pcx-aaaacccc). There is no VPC peering connection between VPC B and VPC C. You cannot route packets directly from VPC B to VPC C through VPC A.

Build Your Own Custome VPC

- VPC and Subnet Sizing The first four IP addresses and the last IP address in each subnet CIDR block are not available for you to use, and cannot be assigned to an instance.
- For Nat Instances you have to disable the Source/Destination Checks.
- Use Nat Gateways instead of Nat Instances

Network Access Control Lists vs Security Groups

- NACL cannot be deployed in multiple VPCs.
- NACL cannot be attached to multiple subnets, only one at the time.
- Each subnet must be associated with a network ACL, if you don't, default NACL will be used.
- By default when you create one NACL, everything is denied.
- Rules are applied in numerical order (starting from the lowest), so when you should create the first rule having number 100 and add others on incremental of 100
- NACL are stateless (opposite of Security Groups)
- Remember to open ephemaral ports on your outbound rules only.
- If you have to block specific IP addresses, use network ACL not Security Groups
- SG apply Layer 5

Custom VPC's and ELB's

- Design tip: Remember that ELB needs at least two public AZ's, so when designing your network, remember to create at least two public subnets in two AZ.

VPC Flow Logs

VPC Flow Logs is a feature that enables you to capture information about the IP traffic going to and from network interfaces in your VPC. Flow log data can be published to Amazon CloudWatch Logs and Amazon S3. After you've created a flow log, you can retrieve and view its data in the chosen destination.

VPC Enpoints

A VPC endpoint enables you to privately connect your VPC to supported AWS services and VPC endpoint services powered by PrivateLink without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection.

- We cannot route traffic to a NAT gateway or VPC gateway endpoints through a VPC peering connection, a VPN connection, or AWS Direct Connect.
- A NAT gateway or VPC gateway endpoints cannot be used by resources on the other side of these connections.
- Conversely, a NAT gateway // VPC gateway endpoints cannot send traffic over VPC endpoints, AWS VPN connections, Direct Connect or VPC Peering connections either.
- Every route table contains a local route for communication within the VPC over IPv4. We cannot modify or delete these routes.
- VPC Endpoints always take precedence over NAT Gateways or Internet Gateways.
- Network ACL rules are evaluated in order, starting with the lowest numbered rule. As soon as a rule matches, it is applied regardless of any higher numbered rule that may contradict it.
- SSH connections are between port 22 of the host and an ephemeral port of the client. In fact, this is true for any TCP service.
- Security groups are stateful, this means any connection initiated successfully will be completed.
- We can create S3 proxy server for enabling use cases where S3 has to be accessed privately through VPN connection, AWS Direct Connect or VPC peering.
- AWS reserves 5 IPs for every subnet, not for every VPC.
- Instances in custom VPCs don't get public DNS hosts by default, we have to set the `enableDnsHostnames` attribute to true. The `enableDnsSupport` is to be set to true too, but that is done by default.
- We can set a custom route table as the main route table.
- We can add secondary CIDR ranges to an existing VPC. When a secondary CIDR block is added to a VPC, a route for that block with target as "local" is automatically added to the route table.
- VPC peering connection route contains Target as `pcx-xxxxxx`. VPN connection // Direct Connect connection route contains Target as `vgw-xxxxxx`.
- VPN is established over a Virtual Private Gateway.
- AWS VPC Endpoints support S3 and DynamoDB. For Amazon ECR, we have to use AWS PrivateLink.
- Difference between DirectConnect and VPN — DirectConnect does not involve the Internet, while VPN does.
- AWS Direct Connect doesn't encrypt in transit data, while VPN does.

To establish a VPN connection, we need —

- A public IP address on the customer gateway for the on-premise network.
- A virtual private gateway attached to the VPC.

To setup AWS VPN CloudHub —

- Each regional site should have non overlapping IP prefixes.
- BGP ASN should be unique at each site.
- If BGP ASN are not unique, additional ALLOW-INs will be required.

The allowed block size in VPC is between a /16 netmask (65,536 IP addresses) and /28 netmask (16 IP addresses).

The following VPC peering connection configurations are not supported —

1. Overlapping CIDR Blocks
2. Transitive Peering
3. Edge to Edge Routing Through a Gateway or Private Connection

We can move part of our on-premise address space to AWS. This is called BYOIP. For this, we have to acquire a ROA, Root Origin Authorization from the the regional internet registry and submit it to Amazon.

- NOTE: Private and public subnets within a VPC can only have one subnet per AZ
- What can you do with a VPC?
    - Launch instances into a subnet of your choosing
    - Assign custom IP address ranges in each subnet
    - Configure route tables between subnets
    - Create single internet gateway and attach it to our VPC
    - Much better security control over your AWS resources
    - Instance security groups
    - Subnet network access control lists (ACLS)
- Default VPC vs Private VPC
    - Default VPC is user friendly, allowing you to immediately deploy instances
    - All subnets have a route to internet
    - No private subnets in default VPC
    - EC2 instance has both a public and private IP address.
- VPC Peering
    - Allows you to connect one VPC with another via a direct network route using private IP addresses
    - Instances behave as if they are on the same private network.
    - You can peer VPCs with other AWS accounts as well as with other VPCs in the same account
    - Peering is in a star config: ie 1 central VPC peers with 4 others. NO TRANSITIVE PEERING!!
- NAT - Network Address Translation
- NAT Instances
    - When creating a NAT instance, Disable Source/Destination Check on the instance.
    - NAT instances must be in a public subnet
    - There must be a route out of the private subnet to the NAT, in order for this to work.
    - The amount of traffic that NAT instances can support depends on the instance size. If you are bottlenecking, increase the instance size.
    - You can create high availability using Autoscaling Groups, multiple subnets in different AZs, and a script to automate failover.
    - Behind a Security Group.
    - Static NAT
    - Dynamic NAT
- NAT Gateways
    - Preferred by the enterprise

    - Scale automatically up to 10G
    - No need to patch OS
    - Not associated with security groups
    - Automatically assigned public IP
    - Must update root tables and point them to NAT Gateway
    - Having one NAT Gateway in one AZ is not good enough, must me redundant in multiple AZs
    - No need to disable Source/Destination Checks
    - More Secure than NAT Instance
- NACL - Network Access Control Lists
    - Can only associate 1 subnet to a Network ACLs
    - Your VPC automatically comes with a default NACL, and by default it allows all inbound and outbound and traffic.
    - You can create custom NACLs. By default each custom NACL denies all inbound and outbound traffic until you add rules
    - Each subnet in your VPC must be associated with a network ACL. If you don't explicitly associate a subnet with a network ACL, the subnet is automatically associated with the default NACL.
    - You can associate a NACL with multiple subnets; however, a subnet can be associated with only one network ACL at a time. When you associate a NACL with a subnet, the previous association is removed.
    - NACLs contain a numbered list of rules that is evaluated in order, starting with the lowest numbered rule
    - NACL have separate inbound and outbound rules, and each rule can either allow or deny traffic.
    - NACL are stateless; responses to allowed inbound traffic are subject to the rules for outbund traffic
    - NACL apply Layyer 4
- ALB
    - You need at least 2 public subnets in order to deeply and application load balancer
- VPC Flow Logs
    - VPC Flow Logs is a feature that enables you to capture info about the IP traffic going to and from network interfaces in your VPC. Flow log data is stored using Amazon Cloudwatch Logs.
    - After you've created a flow log, you can view and retrieve its data in Amazon CloudWatch Logs.
-
- Flow logs can be create at 3 levels
    - VPC
    - Subnet
    - Network Interface Leve
    -

What's IAM

Allows you to manage users and their level of access to the AWS console.

IAM Consist of the following

- Users: People, end users.
- Groups: Collection of users.
- Policies: Are basically the Permissions. Are made up of Documents, formatted in JSON.
- Roles: Is a way to allow one part of AWS to another part.
- Polices (JSON Javascript object notation language)
- Policy
- IAM is a global service
- ARN amazon resource name (arn:aws:sns:us-east-1:428062593321:)

Features

- Centralised control of your AWS account.
- Shared Access to your AWS account.
- Gives you granular Permission.
- Does Identity Federation (Active directory, Facebook, etc)
- MultiFactor Authentication. MFA
- Provides temporary access for users/devices etc.
- Allows you to set up your own password rotation policy.
- Supports PCI DSS compliance.
- integrated with a lot of AWS services.
- explicit deny for new user

IAM Users

- Real identity
- Hard limit 5000 users
- 10 groups for 1 IAM user
- 10 managed policies for 1 IAM user
- character limit 2048

IAM Groups

- Not true identity , only to administra IAM user
- No credentials
- No resource policy refer to the group

IAM Roles

- sts
- Role can not use for login
- Role dont have access keys

- Role assumed

AWS Organizations

- master account can not restricted
- Service control policy for OS

From Console

IAM users sign-in link: You can customize the URL used by users to sign-in

New Users and account set up

- IAM is universal, the region is on a global level.
- Root account: Is the user you used to register on AWS, no one should use it to log-in.
- Always enable MFA on your Root account.
- New Users have No permissions once created.
- Access key ID & Secret are not the same as password.
- You can create and customize your password rotation policies.

There is no default policy ever, anywhere. When permissions are checked, roles and policies are considered together, and in the default case there is no policy, so only the role is considered.

We can configure IAM policies that allows access to specific tags.

Cross-account IAM roles allow customers to securely grant access to AWS resources in their account to a third party.

If our identity store is not compatible with SAML, we can develop a custom application on-premise and use it with STS.

Billing Alarm

You can set a billing alarm to get notified when the amount of spending for the current month reaches your threshold. Before doing it you probably have to go on: https://console.aws.amazon.com/billing/home?#/preferences and enable: Receive Billing Alerts

After that, you can go and use cloudwatch services to set a new Billing Alarm

Cognito and Directory Services

Amazon Cognito has two authentication methods, independent of one another —

- Sign in via third party federation
- Cognito user pools

AWS Directory Service options —

- AWS Managed Microsoft AD

- AD Connector
- Simple AD
- Amazon Cloud Directory
- Amazon Cognito

Connecting AWS SSO to On-Premise Active Directory —

- Two-way trust relationship: Preferred. Users can do everything from both portals.
- AD connector: SSO does not cache user credentials. Users can't reset password from SSO portal, have to do it from on-premise portal.

For two-step verification, SSO sends code to registered email. It can set to be either —

- Always-on
- Context-aware

Microsoft Active Directory supports SAML.

Web console SAML authentication flow

SAML allows configuring federated user access to the corporate AWS account (basically you can re-use your corporate user management system to grant access to AWS and avoid to replicate all your users into IAM).

1. The flow starts from the corporate portal (not on the AWS console)
2. The user triggers the auth on the corporate portal which verifies the user identity
3. The portal generates a SAML authentication response that includes assertions and attributes about the user
4. The client browser is then redirected to the AWS single sign-on endpoint posting the SAML assertion
5. The AWS console endpoint validates the SAML assertion and generates a redirect to access the management console (suing STS)
6. The browser follows the redirect which brings into the AWS console as an authenticated user

STS

Grants users limited and temporary access to AWS resources. Users can come from three sources:

1. Federation (typically Active Directory)
    - Uses Security Assertion Markup Language (SAML)
    - Grants temporary access based off the users Active Directory credentials. This user does not need to be an IAM user.
    - Single sign-on allows users to log in to AWS console without the need of IAM credentials.
2. Federation with mobile apps.
    - Use Facebook/Amazon/Google or other OpenID providers to log in

3. Cross-Account Access.
    - Let's users from one AWS account access resources in another.

Scenario

Typical Authentication Work Flow:

1. Employee enters their username and password
2. The application calls an Identity Broker. The broker captures the username and password.
3. The Identity Broker uses the organization's LDAP directory to validate the employee's identity.
4. The identity Broker calls the new GetFederationToken function using IAM Credentials. The call must include an IAM policy and a duration (1 to 36 hours), along with a policy that specifies the permissions to be granted to the temporary security credentials.
5. The Security Token Service confirms that the policy of the IAM user making the call to GetFederationToken gives permission to create new tokens and then returns four values to the application:
    - Access key
    - secret access key
    - token
    - duration (the token's lifetime)
6. The Identity Broker returns the temporary security credentials to the reporting application.
7. the data storage application uses the temporary security credentials (including the token) to make a request to Amazon S3.
8. Amazon S3 uses IAM to verify that the credentials allowed the requested operation on the given S3 bucket and key.
9. IAM provides S3 with the go-ahead to perform the requested operation.

In the Eaxm

- Scenario 1

1. Develop an Identity Broker to communicate with LDAP and AWS STS.
2. Identity Broker always authenticates with LDAP first, then with AWS STS.
3. The application then gets temporary access to AWS resources.

- Scenario 2

1. Develop an Identity Broker to communicate with LDAP and AWS STS.
2. Identity Broker always authenticates with LDAP first, gets an IAM Role associated with a user.
3. The application then authenticates with STS and assumes that IAM Role.
4. The application uses that IAM role to interact with S3.

Active Directory Integration

- Can you authenticate with Active Directory? ==> Yes, using SAML
- How do you authenticate to get the security credentials into Active Directory? ==> You always authenticate with Active Directory first and then you are assigned the temporary security credentials.

I

The Well Architected Framework

Best Practices

Principle
Authentication
Identity
Authorization

High availability vs Fault tolerance
RTO (Recovery Time Objective) vs RPO (Recovery Point Object)

- Presentation
- Logic
- data

Business Benefits of Cloud

- Almost zero upfront infrastructure cost
- Just-in-time infrastructure
- More efficient resource utilization
- Usage-based costing
- Reduced time to market

Technical Benefits of the Cloud

- Automation: Scriptable infrastructure
- Auto-scaling
- Proactive Scaling
- More efficient development lifecycle
- Improved Testability
- Disaster Recovery and Business Continuity
- "Overflow" the traffic to the cloud

The Well-Architected Framework

The Well-Architected Framework has been developed to help cloud architects build secure, high-performing, resilient, and efficient infrastructure for their applications.

Five pillars:

- Operational excellence
- Security
- Reliability
- Performance efficiency

- Cost optimization

General Design Principles

- Stop guessing your capacity needs.
- Test systems at production scale.
- Automate to make architectural experimentation easier.
- Allow for evolutionary architectures.
- Data-driven architectures.
- Improve through game days.

Pillar 1 Security

Design Principles

- Apply security at all layers.
- Enable traceability.
- Automate responses to security events.
- Focus on securing your system.
- Automate security best practices.
- In transit and at rest encryption

AWS Shared Responsibility Model

Definition

Security in the cloud consists of 4 areas:

Data Protections

Questions you should ask yourself:

- How are you encrypting your data at rest?
- How are you encrypting your data in transit (SSL)?

Privilege Management

Privilege Management ensures that only authorized and authenticated users are able to access your resources, and only in a manner that is intended. It can include:

- Access Control Lists (ACLs).
- Role Based Access Controls.
- Password Management (such as password rotation policies).

Questions you should ask yourself:

- How are you protecting access to and use the AWS root account credentials?

- How are you defining roles and responsibilities of system users to control human access to the AWS Management Console and APIs?
- How are you limiting automated access (such as from applications, scripts, or 3rd party tools or services) to AWS resources?
- How are you managing keys and credentials?

Infrastructure Protection

Questions you should ask yourself:

- How are you enforcing network and host-level boundary protection?
- How are you enforcing AWS service level protection?
- How are you protecting the integrity of the operating system on your AWS EC2 instances?

Detective Controls

You can use detective controls to detect or identify a security breach. AWS Services to achieve this include

- AWS Cloudtrail
- AWS CloudWatch
- AWS Config

Questions you should ask yourself:

- How are you capturing and analyzing your logs?

Key AWS Services

- Data Protection:
  - Encrypt both in transit and at rest using - ELB, EBS, S3 and RDS
- Privilege Management:
  - IAM, MFA
- Infrastructure Protection:
  - VPC
- Detective Controls:
  - AWS Cloud Trail, AWS Config, AWS Cloud Watch

Pillar 2 Reliability

The reliability pillar encompasses the ability of a system to recover from infrastructure or service disruptions, dynamically acquire computing resources to meet demand, and mitigate disruptions such as misconfigurations or transient network issues.

Design Principles

- Test recovery procedures.
- Automatically recover from failure (KPI).
- Scale horizontally to increase aggregate system availability.

- Stop guessing capacity.

Definition

Reliability in the cloud consists of 3 areas:

Foundations

Before architecting any system, you need to make sure you have the prerequisite foundations. With AWS, they handle most of the foundations for you. The cloud is designed to be essentially limitless meaning that AWS handle the networking and compute requirements themselves. However, they do set the service limits to stop customers from accidentally over-provisioning resources.

Questions you should ask yourself:

- How are you managing AWS service limits for your account?
- How are you planning your network topology on AWS?
- Do you have an escalation path to deal with technical issues?

Change Management

You need to be aware of how change affects a system so that you can plan proactively around it. Monitoring allows you to detect any changes to your environment and react.

With AWS, you can use CloudWatch to monitor your environment and services such as autoscaling to automate change in response to your production environment.

Questions you should ask yourself:

- How does your system adapt to changes in demand?
- How are you monitoring AWS resources?
- How are you executing change management?

Failure Management

With cloud, you should always architect your systems with the assumptions that failure will occur. You should become aware of these failures, how they occurred, how to respond to them and then plan on how to prevent these from happening again.

Questions you should ask yourself:

- How are you backing up your data?
- How does your system withstand component failures?
- How are you planning for recovery?

Key AWS Services

- Foundations:
    - IAM, VPC
- Change Management:

        - AWS CloudTrail
    - Failure Management:
        - AWS CloudFormation, AWS RDS multi AZ

Pillar 3 Performance Efficiency

The performance efficiency pillar focuses on the efficient use of computing resources to meet requirements and how to maintain that efficiency as demand changes and technologies evolve. Keep evolving your platform

Design Principles

- Democratize advanced technologies.
- Go global in minutes.
- Use serverless architecture.
- Experiment more often.

Questions you should ask yourself:

- How do you select the appropriate proximity and caching solutions for your system?
- How do you ensure that you continue to have the most appropriate proximity and caching solutions as new solutions are launched?
- How do you monitor your proximity and caching solutions to ensure performance is as expected?
- How do you ensure that the proximity and caching solutions you have matches demand?

Key AWS Services

- Compute
    - Autoscaling
- Storage
    - EBS, S3, Glacier
- Database
    - RDS, DynamoDB, Redshift
- Space-Time Trade-Off
    - CloudFront, ElasticCache, Direct Connect, RDS Read Replicas etc

Pillar 4 Cost Optimization

Use the Cost Optimization pillar to reduce your costs to a minimum and use those savings for other parts of your business. A cost-optimized system allows you to pay the lowest price possible while still achieving your business objectives.

Design Principles

- Transparently attribute expenditure.
- Use managed services to reduce the cost of ownership.
- Trade capital expense for operating expense.
- Benefit from economies of scale.

- Stop spending money on data centre operations.

Definition

Cost optimization in the cloud consists of 4 areas:

Matched supply and demand

Try to optimally align supply with demand. Don't overprovision or under-provision, instead of as demand grows, so should your supply of computing resources. Think of things like Autoscaling which scale with demand.

Similarly in a server-less context, use services such as Lambda that only execute when a request comes in.

Services such as CloudWatch can also help you keep track as to what your demand is.

Questions you should ask yourself:

- How do you make sure your capacity matches but does not substantially exceed what you need?
- How are you optimizing your usage of AWS services?

Cost-effective resources

Using the correct instance type can be key to cost savings. For example, you might have a reporting process that is running on a t2-Micro and it takes 7 hours to complete. That same process could be run on an m4.2xlarge in a manner of minutes. The result remains the same but the t2.micro is more expensive because it ran for longer.

A well-architected system will use the most cost-efficient resources to reach the end business goal

Questions you should ask yourself:

- Have you selected the appropriate resource types to meet your cost targets?
- Have you selected the appropriate pricing model to meet your cost targets?
- Are there managed services (higher-level services than Amazon EC2, Amazon EBS, and Amazon S3) that you can use improve your ROI (return on investment)?

Expenditure Awareness

With cloud, you no longer have to go out and get quotes on physical servers, choose a supplier, have those resources delivered, installed, made available etc. You can provision things within seconds, however, this comes with its own issues.

Many organizations have different teams, each with their own AWS accounts. Being aware of what each team is spending and where is crucial to any well-architected system.

You can use cost allocation tags to track this, billing alerts as well as consolidated billing.

Questions you should ask yourself:

- What access control and procedures do you have in place to govern your AWS costs?
- How are you monitoring usage and spending?

- How do you decommission resources that you no longer need, or stop resources that are temporarily not needed?
- How do you consider data-transfer charges when designing your architecture?

Optimizing Over Time

AWS moves FAST. There are hundreds of new services (and potentially 1000 new services this year). A service that you chose yesterday may not be the best service to be using today.

For example, consider MySQL RDS, Aurora was launched at re:invent 2014 and is now out of preview. Aurora may be a better option now for your business because of its performance and redundancy.

You should keep track of the changes made to AWS and constantly re-evaluate your existing architecture. You can do this by subscribing to AWS blog and by using services such as Trusted Advisor.

Questions you should ask yourself:

- How do you manage and/or consider the adoption of new services?

Key AWS Services

- Matched supply and demand
  - Autoscaling
- Cost-effective resources
  - EC2 (reserved instances), AWS Trusted Advisor
- Expenditure Awareness
  - CloudWatch Alarms, SNS
- Optimizing over time
  - AWS Blog, AWS Trusted Advisor

Pillar 5 Operational Excellence

The Operational Excellence pillar includes operational practices and procedures used to manage production workloads.

Change execution and responses should be automated. All processes and procedures of operational excellence should be documented, tested and regularly reviewed.

Design Principles

- Perform operations with code.
- Align operations processes to business objectives.
- Make regular, small, incremental changes.
- Test for responses to unexpected events.
- Learn from operational events and failures.
- Keep operations procedures current.

Definition

There are three best practice areas of Operational Excellence in the cloud:

Preparation

Effective preparation is required to drive operational excellence. Operations checklists will ensure that workloads are ready for production operation, and prevent unintentional production promotion without effective preparation.

Workloads should have:

- Runbooks: operations guidance that operations teams can refer to so they can perform normal daily tasks.
- Playbooks: guidance for responding to unexpected operational events. Playbooks should include response plans, as well as escalation paths and stakeholder notifications.

In AWS there are several methods, services and features that can be used to support operational readiness and the ability to prepare for normal day-to-day operations as well as unexpected operational events:

- CloudFormation: can be used to ensure that environments contain all required resources when deployed to production and the configuration of the environment is based on tested best practices, which reduces the opportunity for human error.
- Autoscaling or other automated scaling mechanisms: This will allow workloads to automatically respond when business-related events affect operational needs.
- AWS Config: With the AWS Config rules feature create mechanisms to automatically track and respond to changes in your AWS workloads and environments.
  It is also important to use features like tagging to make sure all resources in a workload can be easily identified when needed during operations and responses.
- Document everything:
  Be sure that documentation doesn't become stale or out of date as procedures change.
  Without application designs, environment configs, resource configs, response plans, and mitigation plans, documentation is not complete.
  If documentation is not updated and tested regularly, it will not be useful when unexpected operational events occur. If workloads are not reviewed before production, operations will be affected when undetected issues occur.
  If resources are not documented, when operational events occur, determining how to respond will be more difficult while the correct resources are identified.

Questions you should ask yourself:

- What best practices for cloud operations are your using?
- How are you doing configuration management for your workload?

Operation

Operations should be standardized and manageable on a routine basis. The focus should be on automation, small frequent changes, regular QA testing, and defined mechanisms to track, audit, roll back and review changes. Changes should not be large and infrequent.

In AWS you can set up a continuous integration / continuous deployment (CI/CD) pipeline. Release management processes, whether manual or automated, should be tested and be based on small incremental changes, and tracked

versions. You should be able to revert changes that introduce operational issues without causing any operational impact.

Routine operations, as well as responses to unplanned events, should be automated. Manual processes for deployments, release management, changes and rollbacks should be avoided.

Releases should not be large batches that are done infrequently.

Rollbacks are more difficult in large changes, and failing to have a rollback plan or the ability to mitigate failure impacts will prevent continuity of operations.

Questions you should ask yourself:

- How are you evolving your workload while minimizing the impact of change?
- How do you monitor your workload to ensure it is operating as expected?

Response

Responses to unexpected operational events should be automated. This is not just for alerting but also for mitigation, remediation, rollback and recovery.

Responses should follow a pre-defined playbook that involves stakeholders, the escalation process and procedures. Escalation paths should be defined and include both functional and hierarchical escalation capabilities. Hierarchical escalation should be automated and escalated priority should result in stakeholder notifications.

Questions you should ask yourself:

- How do respond to unplanned operational events?
- How is escalation managed when responding to unplanned operational events?

Key AWS Services

- Preparation: AWS Config provides a detailed inventory of your AWS resources and configuration, and continuously records configuration changes. AWS Service Catalog helps to create a standardized set of service offerings that are aligned to best practices. Designing workloads that use automation with services like AutoScaling, AWS SQS are good methods to ensure continuous operations in the event of unexpected operational events.
- Operations: AWS CodeCommit, AWS CodeDeploy and AWS CodePipeline can be used to manage and automate code changes to AWS workloads. Use AWS SDKs or 3rd party libraries to automate operational changes. Use AWS CloudTrail to audit and track changes made to AWS events.
- Responses: Take advantage of all of the AWS CloudWatch service features for effective and automated responses. CloudWatch alarms can be used to set thresholds for alerting and notification, and CloudWatch events can trigger notifications and automated responses.

SQS

Amazon Simple Queue Service (SQS) is a fully managed message queuing service that enables you to decouple and scale microservices, distributed systems, and serverless applications.

- It's basically a message queue service, like Kafka for example. You add items in the queue, and then someone will ask for these objects from the queue.
- Retention can go up to 14 days.
- It's a pull based system.
- You can have duplicates if you don't manage very well your visibility timeouts.
- Consumers must delete an SQS message manually after it has done processing the message. To delete a message, use the ReceiptHandle of a message, not the MessageId.
- Incoming messages can trigger a lambda function.
- SQS does not encrypt messages by default.
- Default visibility timeout for SQS is 30 seconds.
- Message retention period in SQS — 1 minute to 14 days. The default is 4 days.
- Limit on number of inflight messages — 120,000 for standard queue and 20,000 for FIFO queue.
- Visibility timeout Max 12 Hours

AWS distributed queue service. Type of queses

- **Standard**
    - Extremely high throughput (AWS defines it “nearly unlimited”), highly distributed
    - Messages are guaranteed to be delivered “at-least-once” (sometimes a message can be delivered more than once)
    - Best-effort ordering (messages are generally delivered in order, but they might be occasionally out of order)
- **FIFO (First-In-First-Out)**:
    - Limited number of transactions per second (3000 TPS with batching, 300 TPS without)
    - Guarantees “in-order” delivery and “exactly-once” processing

- **visibility Timeout** (default 30s, min 0, max 12hrs): if a consumer receives a message from the queue, that message is made invisible to other consumers as long as the Visibility Timeout option specifies. This is done to prevent other consumers to pick up the same message again and give some time to the current process to complete the processing and delete the message. If the current consumer crashes before removing the message, once the Visibility Timeout expires, another consumer will be able to pick up the message again.
- **Delay Seconds** (default 0, min 0, max 15min): allows to postpone the delivery of a message. Once added to the queue, a message will remain invisible to consumers for the amount of seconds specified in the Delay Seconds attribute. This attribute can be specified globally at the queue level (in this case the queue is referred as Delay Queue) or at the message level.

Delay queues are similar to visibility timeouts because both features make messages unavailable to consumers for a specific period of time. The difference between the two is that, for delay queues, a message is hidden when it is first added to queue, whereas for visibility timeouts a message is hidden only after it is consumed from the queue

Long polling

Long polling helps reduce the cost of using Amazon SQS by eliminating the number of empty responses

Read the SQS faqs before taking the exam.

SWF

Amazon SWF (Simple Workflow Service) is an Amazon Web Services tool that helps developers coordinate, track and audit multi-step, multi-machine application jobs.

- It ensures that a task is assigned only once and is never duplicated.
- Retention can go up to 1 year
- SWF Workers: Workers are programs that interact with Amazon SWF to get tasks, process received task, and return the result.
- SWF Decider: The decider is a program that controls the coordination of tasks.
- SWF Domains: Domains isolate a set of types, executions, and task lists from others within the same account.
- By default, each workflow execution can run for a maximum of 1 year in Amazon SWF.
- In AWS SWF, a decision task tells the decider the state of the workflow execution.
- SQS vs SWF
- SWF, what does a "domain" refer to? A collection of related workflows
- Workflow Started: An application that can initiate a workflow.
- Decider: Control the flow of activity tasks in a workflow execution.
- Activity Workers: Carry out activity tasks.
-

SNS

Amazon SNS enables message filtering and fan out to a large number of subscribers, including serverless functions, queues, and distributed systems. Additionally, Amazon SNS fans out notifications to end users via mobile push messages, SMS, and email.

- In the exams it will be around Auto Scaling questions.
- SNS stores copies of the messages on multiple AZ.
- With Amazon SNS, there is a possibility of the client receiving duplicate messages
- Push-based
- It can notify through:
    - AWS Lambda functions
    - HTTP/S webhooks
    - mobile push

    - SMS
    - email

Elastic Transcoder

Amazon Elastic Transcoder a is media transcoding service.

API Gateway

Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale.

- You can enable cache
- Scaled automatically
- You can throttle requests
- Connect to CloudWatch to log all requests
- Rest API is statless and any time you access it , it create brand new session

Integration types —

- Lambda function, can be from another AWS account as well.
- HTTP
- Mock
- AWS Service
- VPC Link.

API Gateway Throttling —

- Burst limit refers to the first millisecond.
- Steady-state limit refers to a one second interval.

Throttling behaviors —

- If an user exceeds the burst limit but not the steady-state limit, the rest of the requests are throttled over the one second steady-state interval.
- If an user exceeds the steady-state limit, AWS returns 429 Too Many Requests error.

When it comes to throttling settings, you can override stage settings on an individual method within the stage. That is, there is an option for method level throttling to override stage level throttling.

API Gateway automatically protects the backend systems from DDoS attack.

Cache properties and settings —

Monitoring API Gateway usage — we can use CloudWatch or Access logging. Access logging logs who accessed the API and how the caller accessed the API, CloudWatch does not include this data.

Protect backend systems behind API gateway from traffic spikes —

- Enable throttling.

- Enable result caching.

Kinesis

Amazon Kinesis makes it easy to collect, process, and analyze real-time, streaming data so you can get timely insights and react quickly to new information.

- Kinesis Stream (Data stay)
- Kinesis streams: It consists of shards, where each stream is saved and stored for 24h, up to 7 days. You can have multiple shards on each stream. A consumer then read from the shards and process the data.
- Kinesis Firehose (Data need to process on run time)
- Kinesis Firehose: It doesn't have shards, so data traverses the Firehose. Data has to be processed immediately with lambda or stored in S3 for example.
    - It can stream data to Amazon S3, Amazon Redshift, Amazon Elasticsearch Service, and Splunk.
- Kinesis Analytics: It allows you to analyze the data that exists in Kinesis Firehose of streams.
- Kinesis stream data retention period — 24 hours (default) to 168 hours.
- Amazon Kinesis Scaling Utility is a less cost-effective solution compared to doing it with Cloudwatch alarms + API Gateway + Lambda function

Kinesis data streams store the data, by default for 24 hours and upto 7 days. Whereas Kinesis Firehose stream the data directly into either —

- S3
- Redshift
- Amazon Elasticsearch Service
- Splunk

Kinesis — If ShardIterator expires immediately and data is lost, we have to increase the write capacity assigned to the Shard table.

- Preferred to SQS with cases like order management where there might be manual steps (eg. packing and shipping a parcel) or message delivery needs to be more reliable (only once & in order).
- When a workflow is not making progress is probably because of a missing manual interaction.

Important : If in the exam you got a question which is talking about consuming social media feeds, or a way to consume big data into the cloud, chances are they are talking about

EMR

AWS EMR — AWS Elastic MapReduce, Hadoop based big data analytics.

AWS EMR is preferred for processing log files.

EMR can use spot instances as underlying nodes.

We can access the underlying EC2 instances in AWS EMR cluster.

Important : If in the questions they refer to big data processing, think about Elastic Mapreduce.

KMS and CloudHSM

KMS master keys are region specific.

CloudHSM backup procedure — Ephemeral backup key (EBK) is used to encrypt data and Persistent backup key (PBK) is used to encrypt EBK before saving it to an S3 bucket in the same region as that of AWS CloudHSM cluster.

With AWS CoudHSM, we can control the entire lifecycle around the keys.

AWS KMS API can be used to encrypt data.

OpsWork:

AWS CodePipeline and AWS OpsWorks can't invoke lambda functions.
AWS Opsworks is a configuration management service for Chef and Puppet. With Opsworks Stacks, we can model our application as a stack containing different layers.

AWS STS — The policy of the temporary credentials generated by STS are defined by the intersection of your IAM user policies and the policy that you pass as argument.

AWS VM Import // Export can be used to transfer virtual machines from local infrastructure to AWS and vice-versa.

AWS Trusted Advisor is a resource that helps users with cost management, performance and security.

We can create a CloudTrail log across all regions.

CloudFormation Drift Detection can be used to detect changes in the environment. Drift Detection only checks property values which are explicitly set by stack templates or by specifying template parameters. It does not determine drift for property values which are set by default.

AWS Server Migration Service (SMS) is an agentless service which makes it easier and faster for you to migrate thousands of on-premise workloads to AWS.

AWS Athena is a managed service which can be used to make interactive search queries to S3 data.

Amazon Inspector is a security assessment service, which helps improve security and compliance of applications.

AWS Opsworks is a configuration management service for Chef and Puppet. With Opsworks Stacks, we can model our application as a stack containing different layers.

By default, CloudTrail logs are encrypted using S3 server-side encryption (SSE). We can also choose to encrypt with AWS KMS.

Amazon ECS for Kubernetes (EKS) exists, it's a managed service.

Changes to CloudTrail global service event logs can only be done via the CLI or the SDKs, not the console.

For CloudFront query string forwarding, the parameter names and values used are case sensitive.

AWS Polly — Lexicons are specific to a region. For a single text appearing multiple times, we can create alias using multiple Lexicons.

Amazon Quicksight is a managed service for creating dashboards with data visualization.

AWS Athena pricing is based upon per query and amount of data scanned in each query. To reduce price —

- Partition data based on different parameters so that amount of data scanned gets reduced.
- Create separate workgroups based upon user groups.

AWS CloudSearch helps us add search to our website or application. Like Elasticsearch.

AWS Glue is a fully managed ETL service for data. It keeps a track of processed data using Job Bookmark. Enabling Job Bookmark will help to scan only changes since last bookmark and prevent processing of whole data again.

AWS X-Ray — Helps debug and analyze microservices architecture.

Reducing cost with AWS X-Ray — Sampling at a lower rate.

Amazon WorkDocs has a poweruser facility, which on enabling restricts sharing of documents to that user only.

AWS Data Pipeline can automate the movement and transformation of data for data-driven workflows. For example, transferring older data to S3 from DynamoDB.

Disaster recovery solutions —

- Backup and Restore. Cheapest.
- Pilot Light
- Warm Standby
- Multi-Site
- Multiple AWS Regions. Costliest.

With AWS Config, we can get a snapshot of the current configuration of our AWS account.

For queue based processing, scaling EC2 instances based on the size of the queue is a preferred architecture.

It's best practice to launch Amazon RDS instance outside an Elastic Beanstalk environment.

AWS Athena is simpler and requires less effort to set up than AWS Quicksight.

RI Coverage Budget reports number of instances that are part of Reserved Instance. For an organisation using default IAM policy, each member account owner needs to create a budget policy for individual accounts and not by master account.

Consolidated Billing in AWS Organisations combines usage from all accounts and billing is generated based upon total usage. Services like EC2 and S3 have volume pricing tiers where with more usage volume the overall charge decreases.

To automatically trigger CodePipeline with changes in source S3 bucket, use CloudWatch Events rule and CloudTrail trail.

Amazon Data Lifecycle Manager can be used for creation, retention and deletion of EBS snapshots.

With AWS Organizations, we can centrally manage policies across multiple AWS accounts. With Service Control Policies (SCPs), we can ensure security policies are in place.

AWS WAF is a web application firewall.

In AWS Managed Blockchain network, the format for resource endpoint is — `ResourceID.MemberID.NetworkID.managedblockchain.us-east-1.amazonaws.com:PortNumber`.

When you want to keep your expenditure within a budget, use AWS Budgets, not AWS Cost Explorer.

Cloudwatch monitoring schemes —

- Basic. 5 minutes.
- Detailed. 1 minute.
- Custom. Can be down to 1 second.

Transferring data from an EC2 instance to Amazon S3, Amazon Glacier, Amazon DynamoDB, Amazon SES, Amazon SQS, or Amazon SimpleDB in the same AWS Region has no cost at all.

We can use signed URLs and signed cookies with Cloudfront to protect resources.

Amazon MQ is a message queue which supports industry standard messaging protocols.

Slower login time and 504 errors in front of Cloudfront can be optimized by —

- Lambda @ Edge.
- Setting up an Origin Failover Policy.

AWS Shield is a service that protects resources against DDoS attacks to EC2, ELB, Cloudfront and Route53.

AWS IoT Core is a managed service that lets IoT devices connect and interact with AWS applications and resources.

The following storage have encryption at rest by default —

- AWS Glacier
- Storage Gateway in S3

Perfect Forward Secrecy is supported by —

- Cloudfront
- Elastic Load Balancing

Enabling multiple domains to serve HTTPS over same IP address —- Generate an SSL cert with AWS Certificate Manager and create a Cloudfront distribution. Associate cert with distribution and enable Server Name Indication (SNI).

Classic Load Balancer does not support SNI, we have to use Application Load Balancer or Cloudfront.

The following services enable us to run SQL queries directly against S3 data —

- AWS Athena
- Redshift Spectrum
- S3 Select

By default, each workflow execution can run for a maximum of 1 year in Amazon SWF.

In AWS SWF, a decision task tells the decider the state of the workflow execution.

Third party SSL cert can be imported into —

- AWS Certificate Manager
- IAM Certificate Store

Databases 101

What is a relational DataBase

A relational database is a type of database. It uses a structure that allows us to identify and access data in relation to another piece of data in the database. Often, data in a relational database is organized into tables.

What is a non relational DataBase

A non-relational database is any database that does not follow the relational model provided by traditional relational database management systems. This category of databases also referred to as NoSQL databases, has seen steady adoption growth in recent years with the rise of Big Data applications.

What is data warehousing

Is a system used for reporting and data analysis, and is considered a core component of business intelligence.

OLTP vs. OLAP

We can divide IT systems into transactional (OLTP) and analytical (OLAP). In general, we can assume that OLTP systems provide source data to data warehouses, whereas OLAP systems help to analyze it.

AWS Databases

Relations ones (OLTP)

- SQL
- MySQL
- PostgreSQL
- Oracle
- Aurora
- MariaDB

RDS is not serverless

DyanmoDB - No SQL

RedShift - OLAP

Elasticache - In Memory Caching

Automated Backups

There are two different types of backups for AWS:

- Automated: Are enabled by default, data is **stored in S3 and are enabled automatically**. If you delete the DB, the backup will de belated too
- Database snapshots: Need to be done manually, they are kept even if you remove the DB.

When you restore a backup or a snapshot, the restored version of the database will be in a new RDS instance, with a new DNS name

Encryption You can encrypt your Amazon RDS DB instances and snapshots at rest by enabling the encryption option on your Amazon RDS DB instances.

    - Multi-AZ for disaster recovery (DR) and read repliacas for performance

RDS Multi-AZ

When you provision a Multi-AZ DB Instance, Amazon RDS automatically creates a primary DB Instance and synchronously replicates the data to a standby instance in a different Availability Zone

- it's for disaster recovery not for scaling.

RDS Read Replicas

This feature makes it easy to elastically scale out beyond the capacity constraints of a single DB Instance for read-heavy database workloads The read replica operates as a DB instance that allows only read-only connections

- It's used for scaling not for disaster recovery
- You need to have automatic backups on in order to have a read replica.
- You can have up to 5 read replicas of any DB at the time of writing.
- Each replica has its own DNS name.
- Replicas can become their own databases.
- You can have a replica in another region.
- You can enable encryption on your replica even if your master is not.
- Read replica can be multi A Z
- Read replica increase performance
- Read replica not support SQL server

RDS

- RDS runs on virtual machines (cannot log in; patching is Amazon's responsibility)
- RDS is not serverless
    - Aurora Serverless is serverless
- Automated backups allow you to recover within a retention period between one and 35 days
    - Takes daily snapshots and stores transaction logs which allows for recovery down to a second
    - Enabled by default and stored in S3
- Database snapshots are manual and are still stored if the original RDS has been deleted (unlike automated backups)
- Restored database versions have a new DNS endpoint
- Encryption uses the KMS service for encryption at rest (includes backups, read replicas, and snapshots)
- Read replicas are available for MySQL, PostgreSQL, MariaDB, and Aurora
    - Backups must be enabled
    - Can be in different regions

- Multi-AZ failover can be forced by rebooting the active RDS instance

DynamoDB

Amazon DynamoDB is a key-value and document database that delivers single-digit millisecond performance at any scale. It's fully managed.

- Serverless
- Uses SSD storage.
- Spread across 3 distinct data centres.
- Eventual Consistent Reads (Default)
- Strongly Consistent Reads.
- It's scalable.
- DynamoDB can be very expensive for writes.
- Eventual consistent reads (default) is when consistency across all copies is usually reached within a second (best read performance)
- Strongly consistent reads return a result that reflects all writes that received a successful response prior to the read
- You specify throughput capacity in terms of **read capacity units** and **write capacity units**.
- Eventual Read (It means consistency between all data center . Dynamodb data write in 3 data center thats why)
- Strongly Consistent Reads

Redshift (OLAP)

Amazon Redshift is a fast, scalable data warehouse that makes it simple and cost-effective to analyze all your data across your data warehouse. For Redshift, spot instances are not an option.

To be able to export data into an S3 bucket from a Redshift instance in a private VPC and make sure data doesn't leave VPC:

- Enable Amazon Redshift Enhanced VPC routing
- Create and configure an Amazon S3 VPC endpoint
- Business intelligence
- Three copy of data original , compute node , s3

Redshift stores data by colums.

- Single node: You can start with a single node (Up to 160Gb of ram in one node)
- Multi-Node:
    - Leader node: Manages clients and distribute queries
    - Compute nodes: Store data and execute queries (up to 128 computer nodes max)
- Pricing:
    - You are charged for the total number of hours you run across all your compute nodes.
    - You are charged for backups
    - You are also charged for data transfer within a VPC

        - No charges of leader node
- Security:
    - Encrypted in transit using SSL
    - Encrypted at rest using AES-256 encryption
    - By default redshift takes care of key management.
- Availability:
    - Is available only in 1 AZ
    - Can restore the snapshot to new AZ's in the event of an outage.

Important : If in the question they use languages like business intelligence, or applying business intelligence to big data think about Redshift

Elasticache

Elasticache: Managed, Redis or Memcached-compatible in-memory data store. Basically, it's a DB that saves everything in memory to increases I/O performance.

- Types of Elasticache:
    - Memcached
        - Multi-thread
        - Simple
    - Redis: In memory key-value store
        - advance data type
        - Multi-Az
        - Backup

RDS Aurora

Amazon Aurora is a MySQL and PostgreSQL-compatible relational database built for the cloud, that combines the performance and availability of traditional enterprise databases with the simplicity and cost-effectiveness of open source databases.

- Scaling:
    - Scales in 10GB Increments up to 64TB
    - Compute resources can scale up to 32vCPUs and 244GB of Memory.
    - 2 Copies of your data are contained in each availability zone, with a minimum of 3 availability zones. Total 6 Copies
    - Aurora handles the loss of up two copies of data without affecting database write capability.
    - Aurora handles the loss of up three copies of data without affecting database read capability.
- Replicas:
    - Two types of read replica , aurora replica(Auto failover) and mysql read replica
    - Aurora Replicas: Separate aurora replicas (up to 15 replicas).
    - MySQL Read replicas: (up to 5 replicas).
    - In case of loss of the primary aurora DB, failover will automatically occur on the Aurora replica, it will not in the MySQL read replica.

DNS

What's DNS

The Domain Name Systems (DNS) is the phonebook of the Internet. Humans access information online through domain names, like nytimes.com or espn.com. Web browsers interact through Internet Protocol (IP) addresses. DNS translates domain names to IP addresses so browsers can load Internet resources.

IPv4 vs IPv6

- IPv4 is a 32-Bit IP Address.
- IPv6 is 128 Bit IP Address and was created to fulfil the need for more Internet addresses.

Top Level Domains:

A top-level domain is one of the domains at the highest level in the hierarchical Domain Name System of the Internet (`.com` `.net` `.org` as an example).

In the case of `.co.uk`, `.co` is the second level domain and `.uk` is the top level domain

SOA Record

A Start of Authority record (abbreviated as SOA record) is a type of resource record in the Domain Name System (DNS) containing administrative information about the zone, especially regarding zone transfers.

Example of a SOA record

```
$ dig SOA +multiline google.com

[...]
;; ANSWER SECTION:
google.com.        55 IN SOA ns1.google.com. dns-admin.google.com. (
                238640061  ; serial
                900        ; refresh (15 minutes)
                900        ; retry (15 minutes)
                1800       ; expire (30 minutes)
                60         ; minimum (1 minute)
                )
[...]
```

NS Record

NS stands for 'name server' and this record indicates which DNS server is authoritative for that domain (which server contains the actual DNS records)

```
$ dig NS google.com

[...]
```

```
;; ANSWER SECTION:
google.com.        2073    IN    NS    ns4.google.com.
google.com.        2073    IN    NS    ns1.google.com.
google.com.        2073    IN    NS    ns2.google.com.
google.com.        2073    IN    NS    ns3.google.com.

[...]
```

A Record

The ‘A’ stands for ‘address’ and this is the most fundamental type of DNS record, it indicates the IP address of a given domain

```
$ dig A google.com

[...]

;; ANSWER SECTION:
google.com.        62    IN    A    216.58.201.14

[...]
```

TTL, Time to Live

When a caching (recursive) nameserver queries the authoritative nameserver for a resource record, it will cache that record for the time (in seconds) specified by the TTL.

CNAME Record

CNAME records can be used to alias one name to another. CNAME stands for Canonical Name.

A common example is when you have both example.com and www.example.com pointing to the same application and hosted by the same server.

```
$ dig www.dnsimple.com

[...]

;; ANSWER SECTION:
www.dnsimple.com.    3595    IN    CNAME    dnsimple.com.
dnsimple.com.        55    IN    A    104.245.210.170

[...]
```

CNAME can't be used on the root domain. (This is a contractual limitation imposed by the RFC 1912 and RFC 2181, not a technical one.)

ALIAS record

An ALIAS record is a virtual record type we created to provide CNAME-like behaviour on apex domains. For example, if your domain is example.com and you want it to point to a hostname like myapp.herokuapp.com, you can't use a CNAME record, but you can use an ALIAS record. The ALIAS record will automatically resolve your domain to one or more A records at resolution time, and resolvers see your domain simply as if it had A records.

In AWS you have to use ALIAS records to point your root domain to other DNS records such as your ELB.

DNS IN AWS

Main functions of Route53 —

1. Register domain names.
2. Route internet traffic to the resources for your domain.
3. Check the health of your resources.

It's not used to *distribute* traffic.

Routing policies available in AWS

- Simple routing policy:
  - Only one record with multiple IP addresses
  - Route 53 returns all values in a random order
- Weighted routing policy:
  - Use to route traffic to multiple resources in proportions that you specify. You can send 40% of the traffic on one IP and 60% to another IP.
  - Split traffic based on different weights assigned
  - Set health checks on individual record sets (records that fail are removed until they pass)
- Latency routing policy:
  - Use when you have resources in multiple AWS Regions and you want to route traffic to the region that provides the best latency.
  - Routes traffic based on the lowest network latency for the end user
- Failover routing policy:
  - Use when you want to configure active-passive failover. You need to create a health check before.
  -
- Geolocation routing policy:
  - Use when you want to route traffic based on the location of your users.
- Multivalue answer routing policy:
  - Use when you want Route 53 to respond to DNS queries with up to eight healthy records selected at random.
  - Similar to Simple Routing with health checks to only return values for healthy resources
- Geoproximity routing policy:
  - Use when you want to route traffic based on the location of your resources and, optionally, shift traffic from resources in one location to resources in another.
  - Route traffic based on the geographic location of users and resources
  - Optionally use bias values to route more or less traffic to specific resources

CNAME vs ALIAS —

- For routing to S3 bucket // Elastic load balancer use A record with ALIAS.

- For routing to RDS instance use CNAME with NO ALIAS // without ALIAS.

ALIAS only supports the following services —

- API Gateway
- VPC interface endpoint
- CloudFront distribution
- Elastic Beanstalk environment
- ELB load balancer
- S3 bucket that is configured as a static website
- Another Route 53 record in the same hosted zone

Route53 does not directly log to S3 bucket, we can forward that from Cloudwatch, but can't do it directly.

Types of Route53 health checks —

1. Health checks that monitor an endpoint. This can be on-premise too.
2. Health checks that monitor other health checks.
3. Health checks that monitor Cloudwatch alarms.

Multivalue answer routing policy responds with upto 8 healthy records selected at random.

Weighted routing policy is a good fit for blue-green deployments.